

Machine Learning-Driven Materials Design for Additive Manufacturing Sprint 4

Team 8

Albert Li

Alex Duong

Ryan Calimlim

Introduction

- Pivoted from modelling due to client being unable to obtain training data
- Adding two features:
- Locates pop-in from given load/depth curve
- Generates estimated indentation crack lengths from given image, while allowing user manipulation and real-time length updates

Sprint Objectives

- Add pop-in locator tab to existing program
- Add indentation crack length tab to existing program
- Label training images for indentation crack lengths

AI Usage

- Used AI to compare conventional image processing with ML algorithms for indentation crack length, before peer review led to a decision
- Used AI to figure out best way to add new tab functionality to existing QT GUI

Sprint Achievements and Challenges

- Pop-in tab added, algorithm works without ML
- Training images obtained from client have all been completely labeled and prepped for training
- Program to predict indentation crack lengths has been written with conventional image processing
- Lines can be changed by user and lengths updated in real time per client request
- Needed to adjust to meet new use-case requirements

Next Steps and Retrospective

- Complete ML algorithm to predict crack lengths in indentation images from labeled training images
- Integrate model suggestions into a new crack measurement tab, with adjustable markers and real-time length updates.
- If pop-in training data is obtained, ML algorithm should be created for pop-in as well
- Client feedback established the two-step workflow: suggest measurements, then let researchers review, adjust, or delete them.

Conclusion

- Sprint 4 expanded the application with pop-in detection and indentation crack measurement.
- Added a pop-in tab and developed an image-processing prototype with adjustable measurement lines.
- Completed image labeling to support ML training.
- Next, validate measurement accuracy and integrate ML suggestions while keeping researchers in control of final results.

Thank You!